

From Entropy to Epiplexity

Rethinking Information for Computationally Bounded Intelligence

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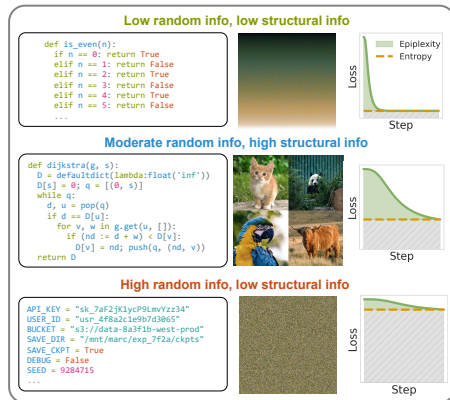
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Speaker: Alex Kravatsky

Which pretraining data support transfer?

- Models pretrained on text can improve performance on robotics, theorem proving and forecasting tasks.
- Dataset size alone does not measure reusable predictive structure.
- **Question:** how much structure can a particular learner extract, given a fixed compute budget?



Training loss on three kinds of data. Dashed line: final loss; green area: loss removed by learning. Source: Fig. 1 of Finzi et al. (2026).

Two classical lenses on information

$H(X) = \mathbb{E}[-\log_2 p(X)]$ is the Shannon entropy of a random variable X with probability mass function p . It is the ideal average code length in bits for a draw from the distribution.

$K(x)$ is the Kolmogorov complexity of one realised string x : the length of the shortest self-delimiting program that outputs x on a fixed universal computer $\mathcal{U}$ (Li and Vitányi, 2008).

Shannon entropy

Uncertainty in a distribution. Uniform n -bit strings have entropy n ; a distribution concentrated on one string has entropy zero.

Kolmogorov complexity

Program length for one string. A repeating pattern can be specified by its rule and length; a typical random string needs about n bits.

Kolmogorov complexity is a mathematical benchmark; no algorithm computes $K(x)$ exactly for every string.

Entropy measures average uncertainty; Kolmogorov complexity measures program length.

The missing ingredient: computational access

Neither classical quantity asks whether a learner can find or use the relevant pattern within a finite budget. For example, a deterministic training run can produce a network with useful circuits, although the rules and random seed form a short description of the run.

- A weak observer may treat a structured process as random.
- A stronger observer may discover a compact predictive model.
- We therefore define information relative to the models available within time T .

Epiplexity measures model bits in the best description available under a time bound.

A model with bounded computation

Fix a prefix-free universal computer $\mathcal{U}$ and a time bound $T(n)$. The observer may use only programs $P \in \mathcal{P}_T$ that, within $T(n)$ steps, can both

- evaluate a probability $P(x)$ for an observation x ;
- sample a new x from the same distribution P .

The program length $|P|$ is measured in bits. This is the idealised definition; in experiments we replace $\mathcal{P}_T$ by models reachable with a stated architecture, optimiser and FLOP budget.

The formal bound limits evaluation and sampling; empirical estimates also account for training.

Epiplexity: the two-part description

Definition 1 (Def. 8 of the paper)

For a random variable X on $\{0, 1\}^n$, let

$$P^* = \arg \min_{P \in \mathcal{P}_T} \{|P| + \mathbb{E}[-\log P(X)]\},$$

with ties broken by the shortest program. Logarithms are base 2. The *epiplexity* of X is $S_T(X) := |P^*|$. The *time-bounded entropy* of X is $H_T(X) := \mathbb{E}[-\log P^*(X)]$.

S_T is the model part: bits needed to describe the selected predictor. H_T is the data part: expected bits needed to encode data with that predictor. Their sum $\text{MDL}_T(X) := S_T(X) + H_T(X)$ is the minimum two-part description length under the bound.

Random variable X	Time bound	$S_T(X)$	$H_T(X)$
U_n , uniform on $\{0, 1\}^n$	enough for uniform model	$O(1)$	$n + O(1)$
0101... or 1010..., probability $\frac{1}{2}$ each	$T(n) = \Theta(n)$	$O(1)$	$O(1)$

For U_n , $S_T = O(1)$ and $H_T \geq n$. For the alternating pair, both parts are $O(1)$.

Two consequences of the time bound

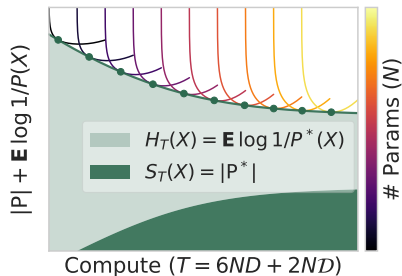
- **Pseudorandom generator.** A short seed is stretched into n bits. Shannon entropy sees only the seed; a polynomial-time observer sees almost n unpredictable bits and almost no structure.
- **One-way map.** For $Y = f(X)$, predicting Y from X is easy, while recovering X from Y is hard. A bounded observer assigns different information to the two orders.

For supervised learning we use the conditional version $S_T(Y | X)$: the input X is available, and only the labels Y are encoded.

The same data can look random or structured depending on what the observer can compute.

From programs to networks: the compute-optimal code

- Search over all programs is intractable, so fix an architecture and optimiser.
- Approximate dense-model accounting:
 $6ND + 2ND \leq T$, where N counts parameters and $D, \mathcal{D}$ count training and evaluation tokens.
- Each run gives a model code plus a validation-data code. Select the shortest admissible two-part code; a lower hull smooths the empirical curve.
- Report its model and data parts as estimates $\hat{S}_T$ and $\hat{H}_T$.

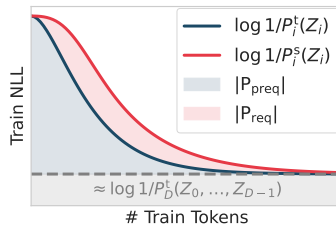


Schematic. Thin curves: training runs, coloured by N . Dots: the lower envelope. Dark green: S_T ; light green: H_T . Source: Fig. 2b of Finzi et al. (2026).

Estimation I: the prequential code

- P_i has seen $Z_0, \dots, Z_{i-1}$; P_0 is the initial model.
- Encode Z_i at ideal cost $-\log_2 P_i(Z_i)$, then update. The decoder repeats the update.
- The sequential data code sums these costs. Subtract the final-model data code to estimate the model part:

$$\hat{K}_{\text{preq}} = \sum_{i < M} [-\log P_i(Z_i) + \log P_M(Z_i)].$$



Dark curve: $\log 1/P_i(Z_i)$ of the network trained on real data (superscript t). Blue area: $|P_{\text{preq}}|$. Red curve and red area: the prequential code of the next slide.
Source: Fig. 2a of Finzi et al. (2026).

- The loss is measured before fitting each example, so it is a test-like loss.
- The area is a heuristic model-part estimate; subtraction alone gives no model-only code.

Estimation II: the requential code

- A teacher model trains on real data; a student model trains on teacher-distributed samples (Qiu et al., 2026).
- Both sides share the student and a random seed. Send the index of a student proposal selected to have the teacher's distribution.
- Relative entropy coding bounds the expected message cost by teacher–student disagreement:

$$\mathbb{E}[\ell_i \mid \mathcal{F}_i] \leq \text{KL}_i + 2 \log(1 + \text{KL}_i) + \kappa, \quad \kappa < 5.21,$$

where $\text{KL}_i = \text{KL}(P_i^t \parallel P_i^s)$ and $\mathcal{F}_i$ is the prior history. Summing gives the student-code bound; $\sum_i \text{KL}_i$ is its leading estimate.

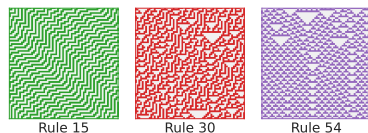
	Prequential	Requential
Model part	heuristic subtraction	explicit student reconstruction
Training	one network	teacher and student, 2–10× slower
Reported values	larger in the paper	smaller in the paper

Prequential is cheaper; requential has an explicit code.

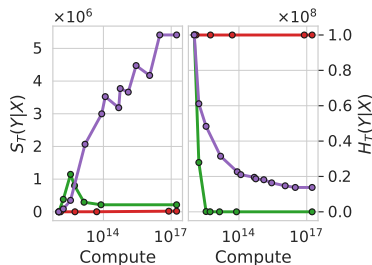
Experiment: cellular automata create structure

An *elementary cellular automaton* (ECA) updates binary cells in parallel from their three-cell neighbourhoods (Wolfram, 2002).

Setting. 64 periodic cells; $Y = F_r^{48}(X)$; transformer sweep; sequential code.



Rule	15	30	54
$S_T(Y X)$, bits	2×10^5	≈ 0	5.4×10^6
$H_T(Y X)$, bits	≈ 0	10^8	1.5×10^7

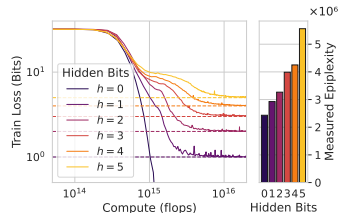
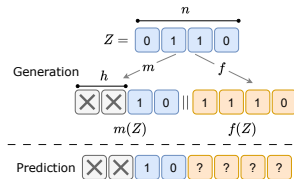


Left: rollouts, time downward. Right: S_T and H_T against compute in FLOPs. Table: our estimates from the plots at 10^{17} FLOPs. Source: Fig. 3 of Finzi et al. (2026).

Rule 15 is simple, rule 30 looks random, and rule 54 contains learnable structure.

Experiment: prediction requires more structure than generation

Setting. Hide h bits of a random $Z \in \{0,1\}^{32}$; predict four rule 30 updates of Z from the visible bits.



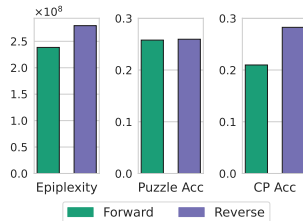
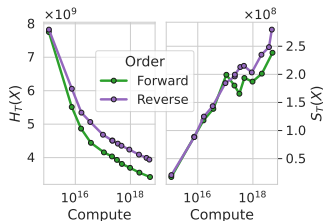
Left: the data. Right: training loss against compute in FLOPs (dashed: h bits) and S_T for each h . Source: Fig. 5 of Finzi et al. (2026).

- The generator runs forward; the predictor must infer the hidden bits.
- The loss approaches h bits only after rapidly growing compute, and S_T grows with h .

Predicting with partial observations requires inference that is absent from forward generation.

Experiment: the order of chess data

Setting. Lichess games in two orders: moves $\rightarrow$ final board, or board $\rightarrow$ moves. Transformer sweep with sequential coding.



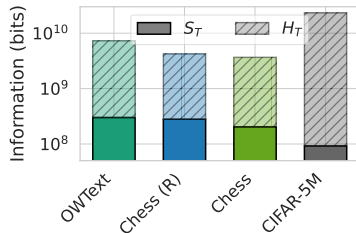
Left: H_T and S_T against compute in FLOPs. Right: S_T and accuracy after fine-tuning. Source: Figs. 4c and 7 of Finzi et al. (2026).

- At large compute, reverse order gives higher H_T and S_T .
- It transfers better to board evaluation: 0.28 vs 0.21 accuracy; puzzles are tied.

The reverse order forces a model of the board state. That structure transfers to a task on board states.

Application: structure in natural data

Setting. Equal token budgets and a shared compute bound. Compare OpenWebText, chess and CIFAR-5M with the sequential code.



Chess (R): reverse order. Source: Fig. 8a of Finzi et al. (2026). Table: our estimates from the figure.

Dataset	S_T , bits	MDL_T , bits	S_T/MDL_T
OpenWebText	3×10^8	7×10^9	4%
Chess, reverse	2.8×10^8	4×10^9	7%
Chess, forward	2×10^8	3.5×10^9	6%
CIFAR-5M	9×10^7	2.3×10^{10}	0.4%

- Under this protocol, text has the largest estimated model part; pixel images have the largest residual part.
- Scaling laws and adaptive data selection show the same ordering (Jiang et al., 2025).

This ranking is relative to the tokenisation, model family and compute budget.

Summary and limitations

Summary

- S_T counts selected model bits; H_T is the expected data-code length.
- A time bound makes computation, prediction order and learned structure observable.
- Estimate S_T with prequential area or teacher–student divergence at the compute-optimal model.

Limitations

- S_T is not a guarantee of downstream usefulness.
- The theorems are asymptotic and cryptographic; the proved lower bound is only $\Omega(\log n)$.
- Estimates depend on architecture, optimiser, T , dataset size and random seed.

Literature

M. Finzi et al. *From Entropy to Epilepsy*. arXiv:2601.03220v2, 2026.

S. Qiu et al. *Requential Coding*. arXiv:2607.11883, 2026.

M. Li, P. Vitányi. *Kolmogorov Complexity*. Springer, 2008.

S. Wolfram. *A New Kind of Science*. 2002.

Y. Jiang et al. Adaptive data optimization. *ICLR*, 2025.

Questions for the audience: answers

1. A dataset X consists of n uniformly random bits. Which quantity is close to n ?

Answer

$H_T(X) \approx n$: the bounded observer cannot predict the bits. A constant-size uniform model gives $S_T(X) = O(1)$.

2. In the prequential estimate, which part of the training-loss curve approximates the epiplexity?

Answer

The area between the loss before each update and the final-model loss, namely $\sum_i [\log 1/P_i(Z_i) - \log 1/P_M(Z_i)]$.

3. What does changing T change?

Answer

It changes the available observer. The total MDL_T cannot increase with T , while S_T itself may rise or fall.

Backup: basic properties

Proposition 1 (Sec. 3 of the paper)

Let c_1, c_2 be constants independent of X . For a random variable X on $\{0, 1\}^n$:

1. $S_T(X) \geq 0$ and $H_T(X) \geq 0$;
2. $H(X) \leq \text{MDL}_T(X) \leq n + c_1$;
3. $\text{MDL}_{T'}(X) \leq \text{MDL}_T(X)$ whenever $T'(n) \geq T(n)$;
4. $\text{MDL}_{T'}(f^{-1}(X)) \leq \text{MDL}_T(X) + |f| + c_2$ with $T'(n) = T(n) + \text{Time}(f)$, for a program f that computes a bijection f in fixed time.

Property 4 is the analogue of $K(f(x)) \leq K(x) + K(f) + c$. Under a time bound a short program for f^{-1} gives no short program for f .

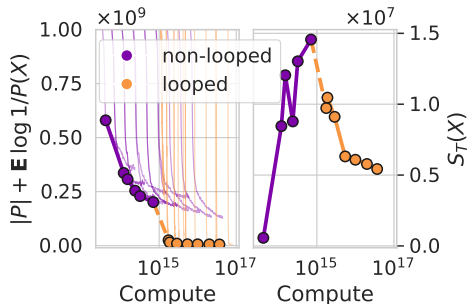
Theorem 1 (Thm. 10 of the paper)

If one-way functions secure against probabilistic polynomial-time algorithms with polynomial advice exist, then some random variables X_n on $\{0, 1\}^n$ have $S_{\text{Poly}}(X_n) = \Omega(\log n)$.

Backup: emergence, where S_T falls as T grows

Setting. ECA rule 54, 64 updates. A non-looped transformer predicts the final state directly. A looped transformer generates the intermediate states one after another: the exact simulation.

- Below 10^{15} FLOPs the non-looped model gives the shorter code. The epiplexity of this model rises to 1.45×10^7 bits: it learns emergent patterns of rule 54.
- Above that bound the looped model gives the shorter code. MDL_T drops to almost zero; S_T falls.
- The authors call this case uncommon. For natural data at moderate compute they expect S_T to grow with compute.



Two-part code and S_T against compute in FLOPs.
Source: Fig. 6 of Finzi et al. (2026).

With less compute the best model is a longer program: S_T need not grow with T .